



Enhancement of piezoelectric properties and temperature stability of high-Curie temperature $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ceramics

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ABSTRACT

Next-generation jet engines, metal forging processes, and non-destructive monitoring of nuclear power plants demand piezoelectric sensors with ultra-high operating temperature, exceptional sensitivity, and outstanding temperature stability. Bismuth calcium niobate ($\text{CaBi}_2\text{Nb}_2\text{O}_9$, CBNO) is considered a candidate for high-temperature piezoelectric materials due to high-Curie temperature (T_C) near 900 °C. However, pure CBNO shows a poor piezoelectric coefficient (d_{33}) and low high-temperature resistivity (ρ). In this work, the piezoelectricity, ferroelectricity, and resistivity of CBNO ceramics were significantly improved by constructing pseudo-tetragonal boundary through the co-substitution of Li/Ce at A site and W/Mo at B site. Remarkably, $\text{Ca}_{0.92}(\text{Li}_{0.5}\text{Ce}_{0.5})_{0.08}\text{Bi}_2\text{Nb}_{1.97}(\text{W}_{2/3}\text{Mo}_{1/3})_{0.03}\text{O}_9$ (CLCBN-3WM) ceramic exhibits the best performance: ultra-high T_C (~ 922 °C), very high d_{33} (~ 16.1 pC/N), a large remanent polarization (~ 11.61 $\mu\text{C}/\text{cm}^2$), and very good high-temperature insulation ρ ($\sim 7.4 \times 10^5$ $\Omega\cdot\text{cm}$ at 600 °C), as well as excellent thermal stability with its d_{33} value degeneration from 16.1 pC/N to 15.1 pC/N from room temperature to 800 °C (less than 7.0 %). These results indicate that CLCBN-3WM ceramics have significant potential for using in electromechanical transducers operating at high temperatures (600 °C or higher).

1. Introduction

Piezoelectric ceramics are widely applied in sensors, actuators, and energy harvesters due to their ability to convert mechanical energy into electrical signals [1–3]. In the past half century, advancements in the nuclear industry, aerospace, and smart materials have introduced more complex operational conditions for piezoelectric ceramics. These challenging conditions have posed difficulties in maintaining stable performance [4]. For example, piezoelectric acceleration transducers in the aerospace sector require ceramics that not only have a high Curie temperature (T_C) but also exhibit excellent piezoelectric coefficients (d_{33}) and high electrical resistivity (ρ) [5–7]. While traditional piezoelectric materials, such as lead magnesium niobate-lead titanate (PMN-PT) [8], bismuth ferrite-barium titanate (BF-BT) [9], and potassium sodium niobate (KNN) [10], exhibited excellent piezoelectric properties, they

present lower T_C (< 500 °C) and poor ρ , rendering them unsuitable for high-temperature applications.

In contrast, Aurivillius-type compounds, also known as bismuth layer-structured ferroelectrics (BLSF), have garnered attention due to their high T_C and outstanding ρ [11]. These properties make them promising candidates for stable and safe operation in high-temperature environments. BLSF ceramics are characterized by the general formula $(\text{Bi}_2\text{O}_2)^{2+}(\text{A}_{n-1}\text{B}_n\text{O}_{3n+1})^{2-}$, where A and B represent metal cations with dodecahedral-coordinated and octahedral-coordinated, respectively. The parameter n denotes the number of $[\text{BO}_6]$ octahedral layers between adjacent bismuth oxide layers, which ranges from 1 to 6 [12–14]. Calcium bismuth niobate ceramics ($\text{CaBi}_2\text{Nb}_2\text{O}_9$, abbreviated as CBNO) are a representative Aurivillius-type compound with $n = 2$. Due to their high T_C (~ 940 °C), substantial spontaneous polarization ($P_s \sim 24.4$ $\mu\text{C}/\text{cm}^2$), and excellent depoling temperature (~ 800 °C), CBNO ceramics have

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garnered significant interest [15,16]. However, the distinctive layered structure of Aurivillius-type compounds restricts the rotation of ferroelectric domains within the a - b plane, thereby making spontaneous polarization along the c -axis challenging [17,18]. As a result, CBNO ceramics exhibit inferior d_{33} and high coercive electric fields, limiting their application in high-temperature piezoelectric sensors [19].

Researchers have employed various approaches, primarily grain orientation techniques and ion doping, to address the suboptimal electrical properties of CBNO-based piezoelectric ceramics. Grain orientation technology involves manipulating the crystal structure orientation within the piezoelectric material to enhance its electrical properties, for instance, the template grain growth (TGG) [20], hot pressing sintering (HPS) [20], and spark plasma sintering (SPS) [21]. However, due to the layered structure of BLSF ceramics and the anisotropy of the crystal, CBNO ceramics can obtain a large piezoelectric coefficient in the texture direction, but the resistance in this direction can deteriorate significantly. Consequently, it is difficult to simultaneously achieve an excellent piezoelectric coefficient and significant resistivity using grain orientation technology. In contrast, ion substitution techniques offer a simple and effective method for enhancing the electrical properties of

CBNO ceramics. Many studies currently focus on co-substituting of rare earth ions and alkali metal ions at the A site of the perovskite-like structure in CBNO ceramics. Modifying the A site to create defect dipoles has proven effective in enhancing the electrical properties and temperature stability of CBNO-based piezoelectric ceramics, such as LiCe [22], NaCe [13], KCe [23], LiBiPr [14], and LiCePr [24]. Moreover, some researchers have explored substituting high-valence ions for Nb in CBNO ceramics, leading to partial crystal structure distortion. Donor doping can introduce excess electrons to compensate for vacancies, thereby improving resistivity, as observed with W [25], WCr [26], and MoCr [27]. Furthermore, it has been demonstrated that chemical doping to construct a pseudo-tetragonal phase structure enhances domain switching and ferroelectric spontaneous polarization in ceramics [28].

In this work, we introduced a co-substitution of Li/Ce at A site and W/Mo at B site into CBNO ceramics. The effects of this co-substitution on the microstructure (including crystal structure and surface morphology) and macroscopic properties (such as dielectric performance, piezoelectric properties, and resistivity) of CBNO-based ceramics has been thoroughly investigated. Fig. 1(a-c) illustrates the effects of ion co-doping on the spontaneous polarization, domain size, and insulation properties of

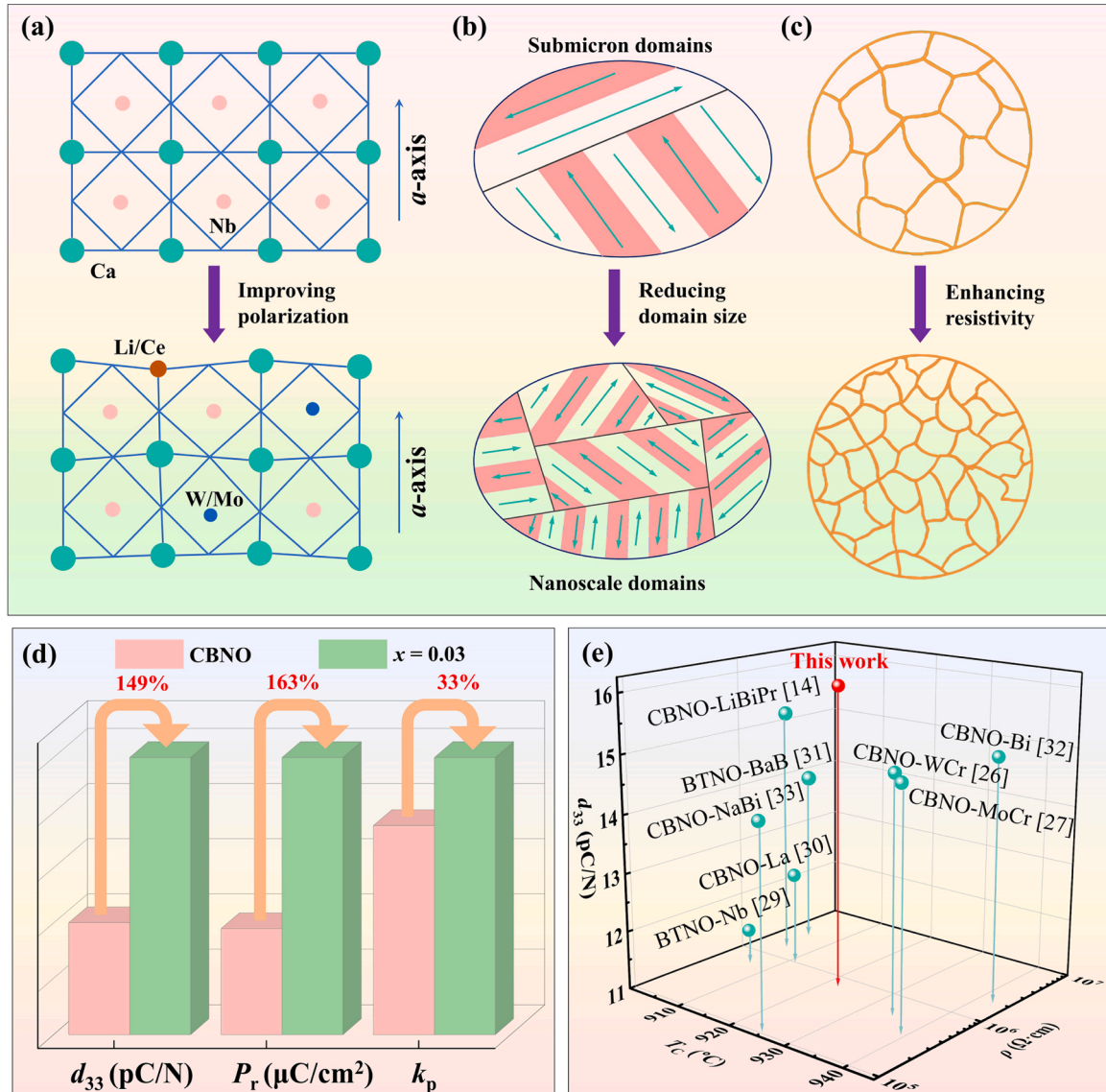


Fig. 1. (a-c) Schematic diagrams of the effect of Li/Ce at A site and W/Mo at B site co-doping on the spontaneous polarization, domain size, and insulating properties of CBNO ceramics, (d) Improvements in d_{33} , P_r , and k_p for $x = 0.03$ ceramic compared to CBNO ceramic, (e) Comparisons in d_{33} , T_c , and ρ (at 600 $^{\circ}\text{C}$) of CLCBN-3WM ceramics and previously reported Aurivillius piezoelectric ceramics.

the ceramics. The composition of CLCBN-3WM ($x = 0.03$) exhibits an enhance of the d_{33} , remanent polarization (P_r), and the planar electro-mechanical coupling coefficient (k_p) over CBNO by approximately 149 %, 163 %, and 33 %, respectively, as shown in Fig. 1(d). The T_C , ρ , and d_{33} of Aurivillius-type high-temperature piezoelectric ceramics ($T_C > 900^\circ\text{C}$) reported in recent years are presented in Fig. 1(e) [14,26,27, 29–33]. Notably, CLCBN-3WM ceramics exhibit a higher d_{33} (~ 16.1 pC/N), which enhances the detection capability of subtle vibrations and shocks, enabling the acceleration sensor to accurately monitor motion states.

2. Experimental

High-temperature piezoelectric ceramics of $\text{Ca}_{0.92}(\text{Li}_{0.5}\text{Ce}_{0.5})_{0.08}\text{Bi}_2\text{Nb}_{2-x}(\text{W}_{2/3}\text{Mo}_{1/3})_x\text{O}_9$ (CLCBN-100xWM, $x = 0, 0.01, 0.02, 0.03, 0.04, 0.05$, and 0.06) were prepared via the solid-state

reaction method. For comparison, undoped CBNO ceramics were also prepared through the same method. All the raw experimental materials were procured from Shanghai Aladdin Biochemical Technology Co., Ltd. The following raw materials were precisely weighed according to the stoichiometric ratios of CLCBN-100xWM: CaCO_3 (99.99 %), Li_2CO_3 (99.99 %), CeO_2 (99.99 %), Bi_2O_3 (99.9 %), Nb_2O_5 (99.9 %), WO_3 (99.99 %), and MoO_3 (99.9 %). Then, ethanol was utilized as the dispersion medium for ball milling for 8 h. The mixed raw materials were subsequently dried and calcined at 850°C for 4 h. The calcined powder was ground again under the same condition. Subsequently, the obtained powder was then dried, ground, and compressed into a disk with a diameter of 8 mm under a pressure of 250 MPa. Finally, through sintering at a temperature of 1100°C for 4 h, the CLCBN-100xWM ceramics were successfully obtained. For samples requiring electrical property testing, both sides will be polished and coated with electrodes made of either silver or platinum.

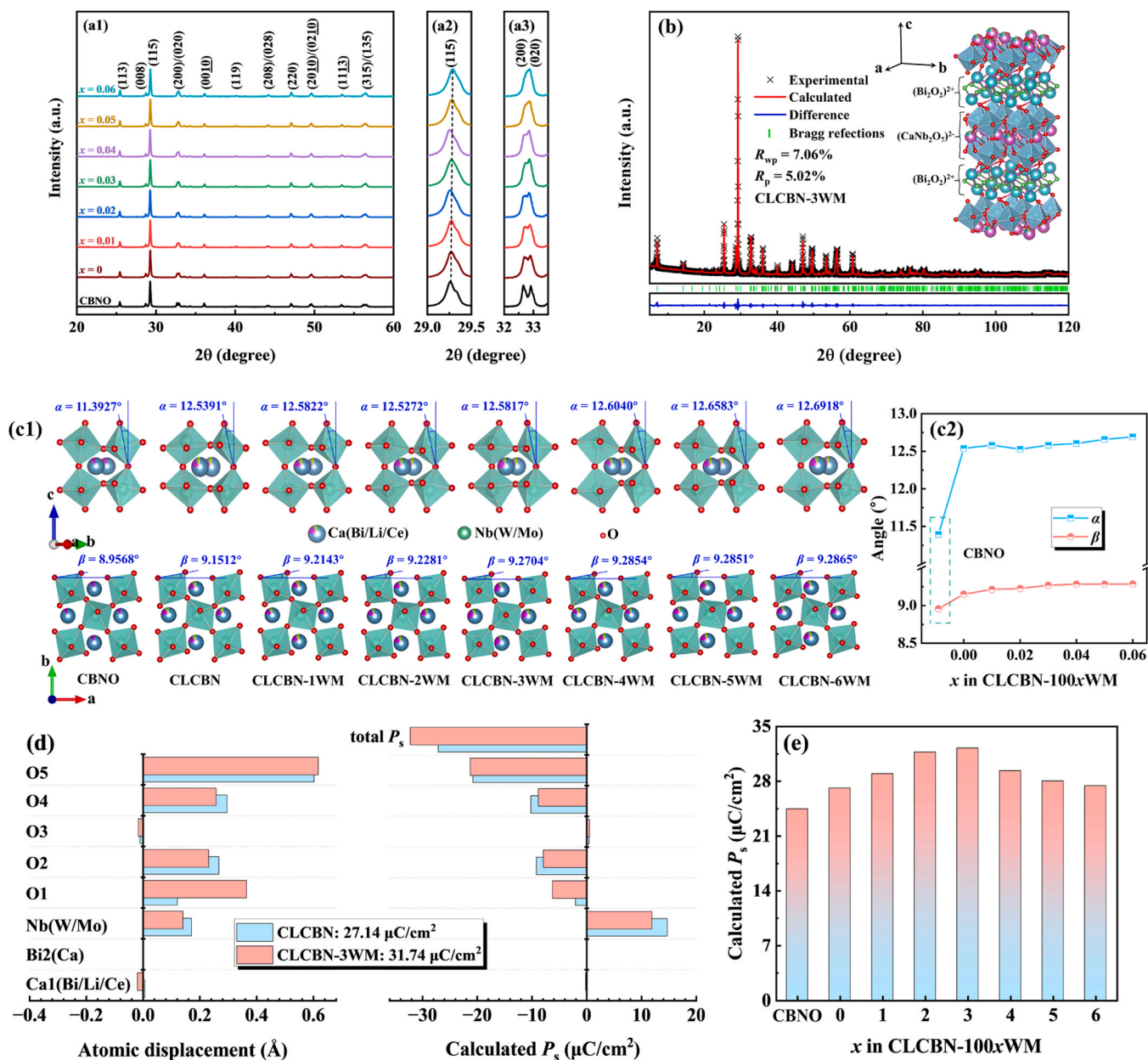


Fig. 2. (a1) XRD patterns of CBNO and CLCBN-100xWM ceramics, (a2) Diagrams in the 2θ range of $29.0\text{--}29.5^\circ$, (a3) Reflections of planes (200)/(020); (b) Rietveld refinement of CLCBN-3WM ceramics; (c1) Visual crystal structures of CBNO and CLCBN-100xWM ceramics in different directions; (c2) Component dependence of α and β ; (d) Atomic displacement of each constituent element and its contribution to P_s in CLCBN and CLCBN-3WM ceramics, (e) Composition dependence of P_s .

The crystal structure and microscopic morphology of CLCBN-100xWM ceramics were characterized by an X-ray diffractometer with $\text{Cu K}\alpha$ radiation (PANalytical Empyrean, PANalytical B.V, Netherlands) and a field emission scanning electron microscope (S-4800, Hitachi, Japan), respectively. Raman spectroscopy data were collected by using a laser confocal micro-Raman spectrometer (DXR, Thermo Scientific, USA) equipped with a 532 nm laser. X-ray photoelectron spectroscopy was conducted with a spectrometer equipped with $\text{Al K}\alpha$ radiation (ESCALAB 250Xi, Thermo Scientific, USA). Dielectric properties data were obtained by an LCR bridge (6500B, Wayne Kerr, UK). Ferroelectric hysteresis loops (P - E) and current density curves (J - E) were measured by using a ferroelectric test system (TF-2000E, Aix ACCT, Germany). Samples were immersed in silicone oil at 180 °C, and an electric field of 150–200 kV/cm was applied for 30 min. The d_{33} value of the polarized samples was measured after discharge by using a quasi-static piezoelectric coefficient meter (ZJ-6BN, Institute of Acoustics, Chinese Academy of Sciences, China). DC resistance and AC impedance spectra were recorded through an electrochemical workstation (VSP, Bio-Logic, France). UV-Vis absorption spectra were collected by a spectroscopy system (UV-3600, Shimadzu Corporation, Japan). Domain structure and high-resolution transmission electron microscopy (HR-TEM) were obtained using a transmission electron microscope (JEM-2100F, JEOL, Japan).

3. Results and discussion

Room-temperature XRD patterns of both CBNO and CLCBN-100xWM ceramics are presented in Fig. 2(a1), all reflection peaks match the CBNO standard card (PDF # 49-0608, $A2_1am$ space group), without second phase material being generated. This suggests that Li, Ce, W, and Mo can be incorporated into the CBNO lattice, forming a solid solution within the designed doping range. The (115) reflection exhibits the highest intensity, this is consistent with the strongest reflection of (112 n +1) in BLSF ceramics [27,34]. The detailed contours near 29.0° and 29.5° are illustrated in Fig. 2(a2). Notably, the reflection of (115) gradually shifts to a higher angle with increasing W/Mo doping, attributed to the replacement of larger Nb^{5+} ion (0.64 Å, CN = 6) by small W^{6+} ion (0.60 Å, CN = 6) and Mo^{6+} ion (0.59 Å, CN = 6), leading to unit cell contraction [35]. Additionally, the distance between the (200) and (020) reflections is close to zero with increased W/Mo doping, as displayed in Fig. 2(a3). The reflections (200) and (020) merge into a single peak at doping levels greater than 0.03, indicating an evolution of the crystal structure of the ceramic from the orthorhombic phase to a pseudo-tetragonal phase. This suggests a pseudo-tetragonal phase boundary near $x = 0.03$, aligning with previously reported findings for CBNO-based piezoelectric ceramics [13,28,36]. For bismuth layer-structured piezoelectric ceramics, the pseudo-tetragonal structure is known to enhance electrical performance.

To further analyze the crystal structure evolution of CLCBN-100xWM ceramics, we performed Rietveld refinement on the collected powder XRD data based on the structural model (space group: $A2_1am$) reported by Blake *et al.* [37], as shown in Fig. S1 and Table S1. The calculated patterns correspond well with the experimental values, exhibiting reasonable χ^2 , R_p , and R_{wp} values. It is observed that after W/Mo doping, the lattice parameters a and c decrease, while b increases. The ratio of a/b reflects the change in tetragonality. The a/b value decreases from 1.00539 to 1.00395 with increasing doping in CLCBN-100xWM ceramics, indicating a gradual deviation of the crystal structure from the orthorhombic phase and a trend toward a pseudo-tetragonal phase with higher symmetry. Fig. 2(b) presents the crystal structure diagram of CLCBN-3WM ceramics. The refined atomic occupancy indicates cation disorder between the $(\text{CaNb}_2\text{O}_7)^{2-}$ layer and the $(\text{Bi}_2\text{O}_2)^{2+}$ layer. The occupancy of Ca in the perovskite A site and the $(\text{Bi}_2\text{O}_2)^{2+}$ layer is 0.795 (7) and 0.125(7), respectively. Similarly, the occupancy of Bi in the perovskite A site and the $(\text{Bi}_2\text{O}_2)^{2+}$ layer is 0.053(2) and 0.947(2), respectively.

In Aurivillius-type compounds, the contribution to spontaneous polarization (P_s) in the crystal structure primarily arises from the rotation of the oxygen octahedron along the c -axis and the tilt of the a - b two-dimensional plane in the perovskite-like layer [38]. The degrees of tilt and rotation of the $[\text{NbO}_6]$ octahedron are represented by α (the angle between O1-O2 and the c -axis) and β ($\angle\text{O4-O4-O4}$), respectively. The values of α and β for all samples were calculated based on atomic occupancies obtained from Rietveld refinement, as illustrated in Fig. 2(c1). It is observed that as the doping amount increases, both α and β gradually increase. Specifically, the variation of α is significantly greater than that of β , as shown in Fig. 2(c2), indicating that the rotation of the $[\text{NbO}_6]$ octahedron has a more significant effect on the spontaneous polarization intensity in CLCBN-100xWM ceramics.

To elucidate the contribution of each component ion to the spontaneous polarization (P_s) of ferroelectrics, we employ the spontaneous polarization intensity calculation model for displacement ferroelectrics proposed by Shimakawa to calculate the P_s of the sample [39]. This model is based on the atomic coordinates finely adjusted by the Rietveld method, as listed in Tables S2-S6.

$$P_s = \sum_i \frac{m_i \times \Delta x_i \times Q_i e}{V} \quad (1)$$

where m_i represents the site multiplicity, Δx_i is the displacement of the atom along the a -axis from its corresponding position in the parent structure, $Q_i e$ is the charge carried by the ion, and V represents the volume of a unit cell. The atomic displacement and contribution to P_s of each constituent element of CLCBN and CLCBN-3WM ceramics are illustrated in Fig. 2(d). CLCBN-3WM ceramics exhibit a large P_s (31.74 $\mu\text{C}/\text{cm}^2$), which is primarily attributed to the larger atomic displacement of O5. In addition, the composition dependence of P_s is presented in Fig. 2(e). The P_s of CLCBN-100xWM ceramics are larger than that of CBNO, which may be due to the increase in lattice distortion caused by the higher doping amount, thus further affecting P_s . For $x > 0.03$ components, P_s shows a downward trend, possibly due to the excessive deviation of the ceramic crystal structure from the parent phase, which increases the tetragonality and results in a decrease in P_s .

To investigate the effect of the W/Mo doping amount on molecular vibration in ceramic crystals, Raman spectroscopy was conducted on all samples, and the results are presented in Fig. S2(a). All samples exhibit Raman vibration modes consistent with the orthorhombic phase structure within the tested range (20–1000 cm^{-1}). Notably, strong phonon vibration modes are observed at approximately 36.5, 55.7, 179.2, 201.3, 583.6, and 815.2 cm^{-1} , denoted as ν_1 - ν_6 , respectively. Generally, the low-wavenumber vibration mode below 200 cm^{-1} is primarily associated with the A site ions in the perovskite structure, while the high-wavenumber vibration above 200 cm^{-1} is attributed to the $[\text{NbO}_6]$ octahedron. By employing the Lorentz function decomposition of the Raman vibration modes, it is possible to extract 16 independent vibration modes. Fig. S2(b) illustrates the ν_1 - ν_6 Raman peaks in CLCBN-100xWM ceramics. Among these peaks, ν_1 - ν_3 are associated with A site Ca^{2+} ions, and ν_1 and ν_3 almost do not shift with the increase in doping amount, while ν_2 experiences a decrease in wavenumber compared to CBNO, attributed to the substitution of Li^+ and Ce^{3+} for Ca^{2+} , altering the original Ca^{2+} ion vibration. The ν_4 peak reflects the degree of distortion in the $[\text{NbO}_6]$ octahedron. ν_5 and ν_6 represent the asymmetric and symmetric vibrations of the $[\text{NbO}_6]$ octahedron, respectively. In CLCBN-100xWM ceramics, both ν_5 and ν_6 show a shift towards lower wavenumbers compared to CBNO. This can be attributed to the rotation and tilt of the octahedron caused by the doping of W/Mo composite ions at the B site, resulting in localized lattice structure distortion [12].

Fig. 3(a-h) presents the surface morphology of CBNO and CLCBN-100xWM ceramics. Generally, all samples exhibit rod-like or long strip-like grains with a large aspect ratio and anisotropic growth, which can be attributed to the structure of bismuth layered materials. The lack of noticeable pores suggests that these ceramics have good insulation

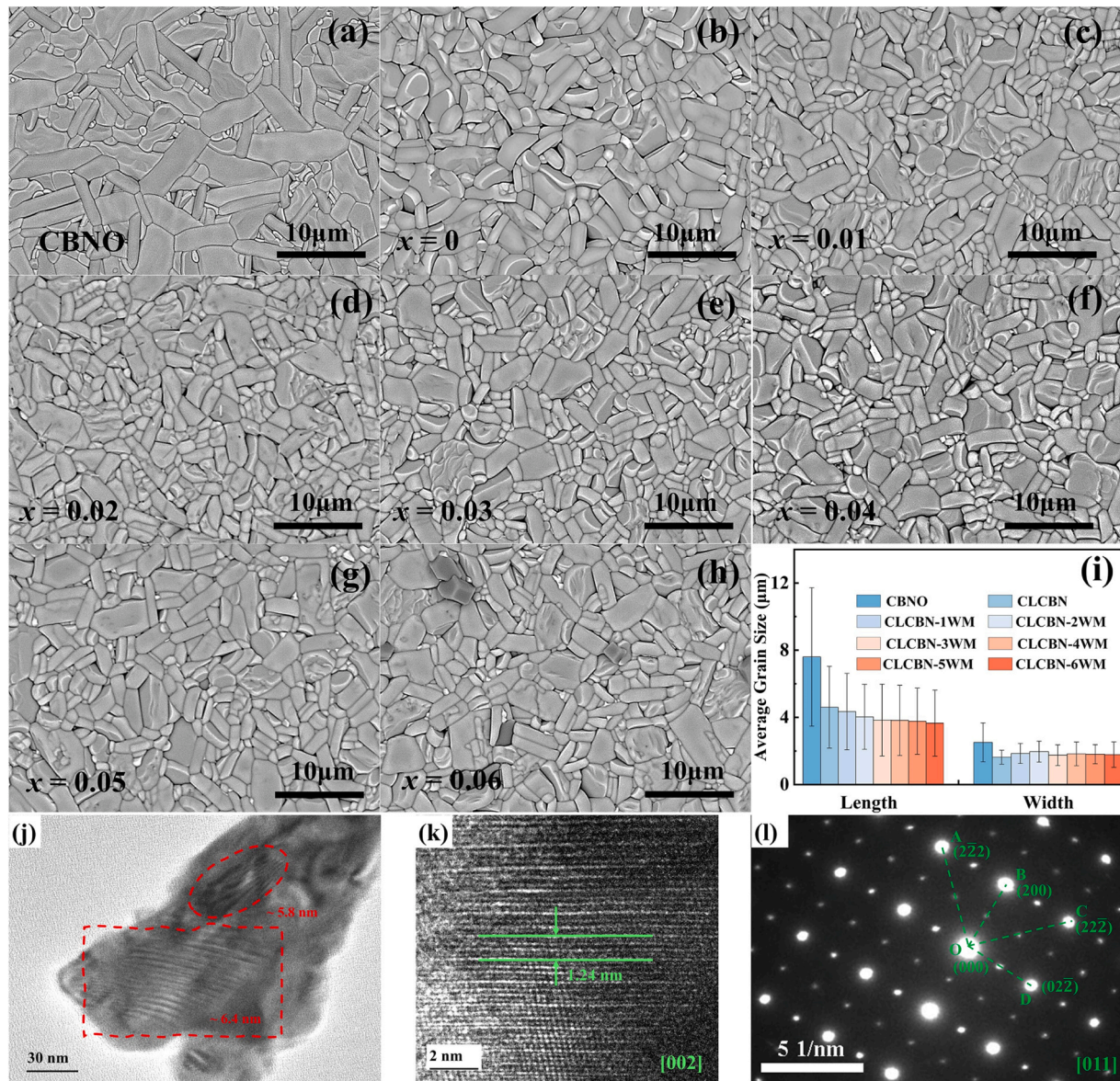


Fig. 3. Surface microstructure of CBNO (a) and CLCBN-100xWM (b-h) ceramics, (i) Average grain size varies with the composition. Striped domain structure (j) HR-TEM (k), and SAED (l) images in CLCBN-3WM ceramics.

properties. Fig. 3(i) illustrates the distribution of average grain size (length and width) for all samples. The average grain size decreases significantly from 7.06 μm (length) to 4.61 μm (length) after doping with Li/Ce at the A site. This indicates that appropriate doping with Li/Ce effectively refines the ceramic grains, which is consistent with previous research conclusions [38,40]. The growth of CLCBN-100xWM ceramic grains in the a - b direction is inhibited, and the grains transform from rod-like structures to block-like forms. The average grain size decreases gradually from 4.61 μm (length) to 3.66 μm (length) with the increase of W/Mo co-doping, accompanied by a gradual reduction in the aspect ratio (length/width, as shown in Fig. S3). This phenomenon may be attributed to the decreased oxygen vacancy concentration in CLCBN-100xWM ceramics (as discussed later), which impedes ion diffusion during the densification process, thereby slowing both grain and grain boundary diffusion rates and inhibiting overall grain growth [41]. Notably, the microstructure of CLCBN-6WM ceramics includes a small number of abnormal diamond-shaped grains alongside block and elongated strip grains. Energy dispersive spectroscopy (EDS) mapping (Fig. S4) reveals that the diamond-shaped grains in CLCBN-6WM ceramics are enriched with Ca, W, and Mo elements. In addition, we

synthesized a sample with $x = 0.07$ for XRD analysis and found the formation of impurity phases, as shown in Fig. S5. It can be inferred that the solid solution limit of W and Mo in the CLCBN-100xWM system is approximately $x = 0.06$.

The domain structure of the CLCBN-3WM ceramic sample was further elucidated through TEM images, as depicted in Fig. 3(j). The images reveal a rich nano-scale 180° domain structure with a width of approximately 6 nm. Previous studies on CBNO ceramics using PFM amplitude imaging showed with a width of around 200 nm [22]. The reduction in domain size and increase in domain wall density are beneficial for enhancing the piezoelectric properties and temperature stability of CBNO-based piezoelectric ceramics [32]. Fig. 3(k) presents HR-TEM images of CLCBN-3WM ceramics, revealing periodic stripes associated with the bismuth oxide layer and $[\text{NbO}_6]$ layer. The measured distance between adjacent stripes is approximately 1.24 nm, which is about half of the lattice parameter c of CLCBN-3WM ceramic ($c = 2.48619$ nm, as shown in Table S1). Fig. 3(l) displays the selected area electron diffraction (SAED) images of CLCBN-3WM ceramics along the $[011]$ axis. Weak $(hk0)$ and $(h00)$ type superlattice reflections are observed in the images, a phenomenon previously reported by Pan et al.

[42]. Additionally, a similar superlattice phenomenon has been noted in $\text{SrBi}_2\text{Nb}_2\text{O}_9$ ceramics [43].

The frequency dependence of the dielectric permittivity (ϵ') and dielectric loss ($\tan\delta$) for CLCBN and CLCBN-100xWM ceramics is shown in Fig. S6(a-h). After polarization by a DC electric field, the dielectric permittivity of all samples decreases, which may be due to a reduction in non-180° domain wall density or changes in domain structure [44]. The resonance and anti-resonance peaks observed at high frequencies in the polarized samples can be attributed to the piezoelectric effect. Fig. S6(i) illustrates the composition dependence of the dielectric permittivity and dielectric loss for all samples at 100 kHz. Dielectric permittivity improves after doping, which is attributed to the decrease in oxygen vacancy concentration [19]. However, after DC polarization, the dielectric loss of the CLCBN-100xWM ceramic sample increases, which can be attributed to changes in dielectric loss caused by the movement and rearrangement of the domain wall under the electric field [19]. Additionally, charge migration or accumulation within the ceramic can lead to an uneven distribution of the local electric field intensity during the polarization process, which in turn affects the dielectric properties of the ceramic.

The temperature dependence of the dielectric permittivity and dielectric loss of all samples at 1 MHz is shown in Fig. 4(a-b). All samples exhibit a maximum dielectric permittivity due to the ferroelectric to paraelectric phase transition, corresponding to the Curie temperature (T_C). As the doping content increases, the T_C of the sample decreases from 940 °C to 914 °C, which may be attributed to the enhanced tetragonality of the ceramic crystal structure and a reduced energy barrier for phase transition due to the increased doping content.

The distortion of the perovskite structure in Aurivillius-type compounds can also be estimated by the tolerance factor (t) [17]:

$$t = \frac{R_A + R_O}{\sqrt{2}(R_B + R_O)} \quad (2)$$

where R_A , R_B , and R_O represent the ionic radii of A site ions, B site ions, and oxygen ions, respectively. In CLCBN-100xWM ceramics, the position of Nb is replaced by W/Mo with smaller ionic radii. Consequently, the value of t increases with higher doping levels of W/Mo. In generally, a larger t is often associated with a lower T_C [45], which is consistent with the experimental measurements. Additionally, the trend in T_C corresponds with the orthorhombicity parameter $2(a-b)/(a+b)$, indicating that T_C is influenced by the degree of crystal distortion.

The ferroelectric hysteresis loops (P - E) and current density curves (J - E) of CBNO and CLCBN-100xWM ceramics, as illustrated in Fig. 5(a-h). The J - E curves of CLCBN-100xWM ceramic samples show pronounced current peaks. CLCBN-3WM demonstrates the most standard P - E loops. In contrast, the P - E loops for components with $x > 0.03$ gradually

narrow, which can be attributed to three reasons. Firstly, the reduction in grain size increases grain boundary density and lattice stress, thereby impeding domain switching. Secondly, a higher level of substitution enhances lattice distortion and alters crystal symmetry. Thirdly, excessive substitution affects interactions between the bismuth oxide layer and the oxygen octahedral layer, leading to lattice mismatch and a decline in ferroelectric properties. The asymmetry observed in the P - E loops for $x > 0.03$ is attributed to the bias electric field within the ceramic, caused by the pinning effect of defect dipoles and the decreased grain size [46,47]. The variation of remanent polarization (P_r) with composition is illustrated in Fig. 5(i). The P_r initially increases and then decreases, with CLCBN-3WM ceramics achieving an optimal value ($\sim 11.61 \mu\text{C}/\text{cm}^2$), which may be related to its large spontaneous polarization (P_s). Therefore, an appropriate amount of W/Mo co-substitution for Nb can enhance ferroelectric properties.

Fig. 6(a) illustrates the variations in piezoelectric coefficient (d_{33}) and planar electromechanical coupling coefficient (k_p) of CLCBN-100xWM ceramics with different doping levels. Both d_{33} and k_p initially increase and then decrease with the doping amount increases, showing a trend consistent with P_r . The optimal values are achieved at $x = 0.03$ ($d_{33} \sim 16.1 \text{ pC/N}$, $k_p \sim 7.7\%$). Notably, d_{33} is roughly 2.5 times higher than that of CBNO ($\sim 6.5 \text{ pC/N}$), which may be attributed to the presence of a pseudo-tetragonal distortion. The piezoelectric properties of Aurivillius-type compounds primarily originate from the contributions of the perovskite-like layers. Eq. (3) can be used to evaluate the piezoelectric properties of CLCBN-100xWM ceramics [47]:

$$d_{33} = 2Q\epsilon_0\epsilon'P_r \quad (3)$$

where Q and ϵ_0 denote the electrostrictive coefficient and the dielectric permittivity in vacuum, respectively. For perovskite-type ferroelectrics below T_C , the Q value remains relatively stable with changes in composition and temperature. Consequently, d_{33} can be determined by the product of ϵ' and P_r . The trend in the variation of ϵ' and P_r aligns with that of d_{33} [as shown in Fig. 6(a)]. Hence, the CLCBN-3WM ceramics, characterized by larger ϵ' and P_r values, exhibit superior piezoelectric properties.

The thermal depoling behavior of CLCBN-100xWM ceramics was evaluated by measuring the d_{33} values of the samples after annealing at different temperatures (25°C $\sim$ 940 °C) for 2 h, as shown in Fig. 6(b). The results indicate that the d_{33} value for all CLCBN-100xWM samples decreases slightly at low temperatures (< 400 °C), which is attributed to the presence of unstable non-180° domains within the samples [48]. As the temperature increases, these domains revert to their initial state due to internal ceramic stress, causing a slight decrease in d_{33} . Fig. 6(c) illustrates the relationship between normalized d_{33} and annealing temperature, revealing that an appropriate amount of W/Mo co-substitution

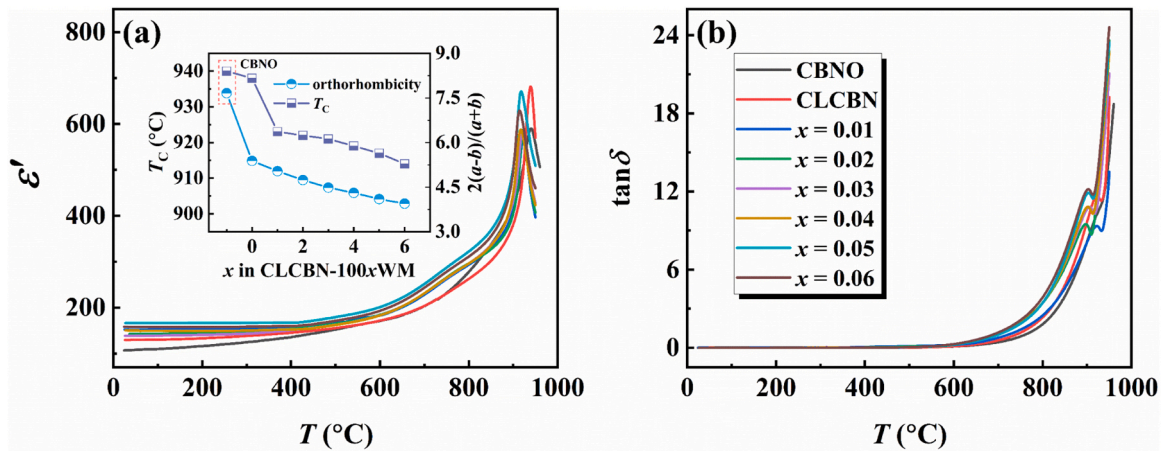


Fig. 4. Temperature dependence of dielectric permittivity (a) and dielectric loss (b) of CBNO and CLCBN-100xWM ceramics at 1 MHz.

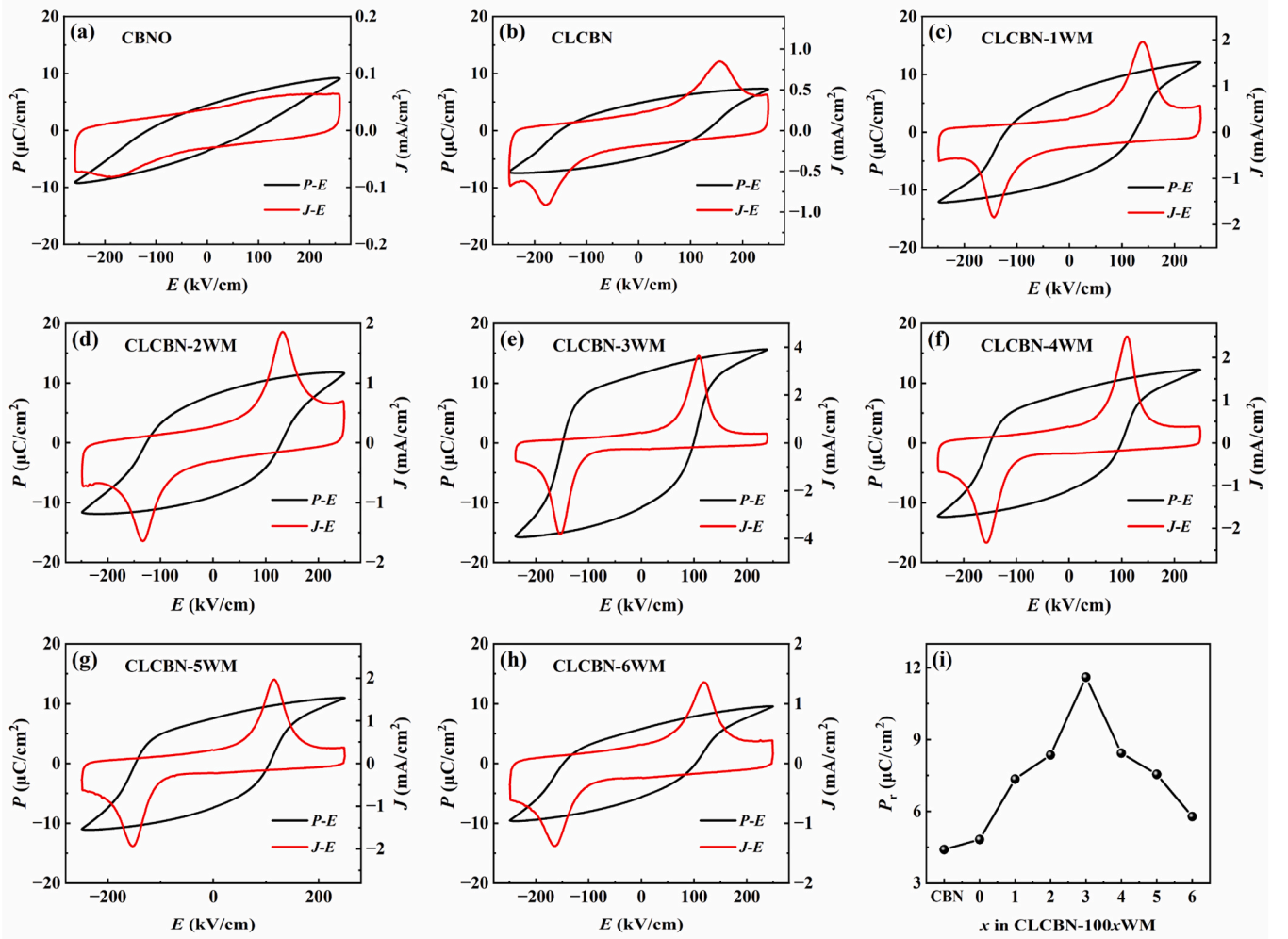


Fig. 5. (a-h) P - E and J - E loops of all samples measured at 160 °C, 250 kV/cm, and 10 Hz, (i) The composition dependence of P_r .

for Nb can improve the d_{33} stability, which is caused by the decrease in oxygen vacancy concentration. Fig. 6(d) shows the variation of normalized d_{33} with composition after annealing at 800 °C for 2 h. It is evident that $d_{33}^{800\text{ °C}}/d_{33}^{\text{RT}}$ initially increases and then decreases with the rise of x , possibly due to the pseudo-tetragonal structure. Liu *et al.* have confirmed that an appropriate amount of the pseudo-tetragonal phase enhances the thermal depoling behavior of CBNO-based ceramics [13]. Additionally, reduced grain sizes increase the energy barrier for domain wall movement, complicating the switching of 180° domains and enhancing the thermal stability of ceramics. Specifically, the ceramic with $x = 0.03$ exhibits outstanding thermal stability, retaining 93.7 % (~ 15.1 pC/N) of the room temperature d_{33} (~ 16.1 pC/N).

The elemental composition and chemical valence states of CBNO, CLCBN, CLCBN-3WM, and CLCBN-6WM ceramics were assessed using XPS analysis, as illustrated in Fig. S7(a). The presence of peaks corresponding to Ca, Bi, Nb, O, and C indicates that these elements are present in the ceramics, with additional C 1 s peaks appearing during the testing process. High-resolution XPS spectra of the O 1 s orbitals are presented in Fig. S7(b). The O 1 s spectrum can be deconvoluted using a Gaussian function into four distinct modes: lattice oxygen (O^{2-}) at approximately 529.91 eV, highly oxidized oxygen species ($\text{O}_2^{2-}/\text{O}^-$) at 530.88 eV, carbonate or C-O bond (CO_3^{2-} or C-O) at 532.07 eV, and molecular water or hydroxyl (H_2O or -OH) due to physical adsorption at 533.5 eV [49,50]. Previous research has demonstrated a close relationship between $\text{O}_2^{2-}/\text{O}^-$ and the concentration of surface oxygen vacancies [51–53]. Fig. S7(c) presents the relative area percentage of $\text{O}_2^{2-}/\text{O}^-$ in the estimated O 1 s

fitting peak. It is evident that compared to CBNO ceramics, the relative proportion of $\text{O}_2^{2-}/\text{O}^-$ in CLCBN ceramics is higher, indicating a greater concentration of oxygen vacancies. However, the area of $\text{O}_2^{2-}/\text{O}^-$ in CLCBN-3WM and CLCBN-6WM ceramics is relatively small, suggesting that W/Mo doping can effectively reduce the concentration of oxygen vacancies in the ceramics.

Fig. 7(a) illustrates the temperature dependence of resistivity for CLCBN-100xWM ceramics. ρ gradually decreases with the temperature increases, demonstrating a negative temperature coefficient characteristic of the CLCBN-100xWM sample. Moreover, resistivity initially increases and then decreases with increasing doping levels. The resistivity of the $x = 0.03$ component is the largest ($\sim 7.4 \times 10^5 \Omega\cdot\text{cm}$ at 600 °C), approximately seven times that of CBNO ($\sim 1.1 \times 10^5 \Omega\cdot\text{cm}$ at 600 °C). The increase in resistivity for components with $x < 0.03$ may be attributed to the introduction of W/Mo, which increases atomic diversity in the material. These atoms can interact with carriers, leading to carrier scattering that reduces carrier migration efficiency and raises resistivity. Furthermore, donor doping generates excess free electrons that can counteract the oxygen vacancy ($\text{V}_\text{O}^{\bullet\bullet}$) and holes ($\text{h}^{\bullet}$) produced by Bi_2O_3 volatilization at high temperatures. This compensation decreases oxygen vacancy concentration, thereby increasing resistivity. The defect reaction equation is as follows:



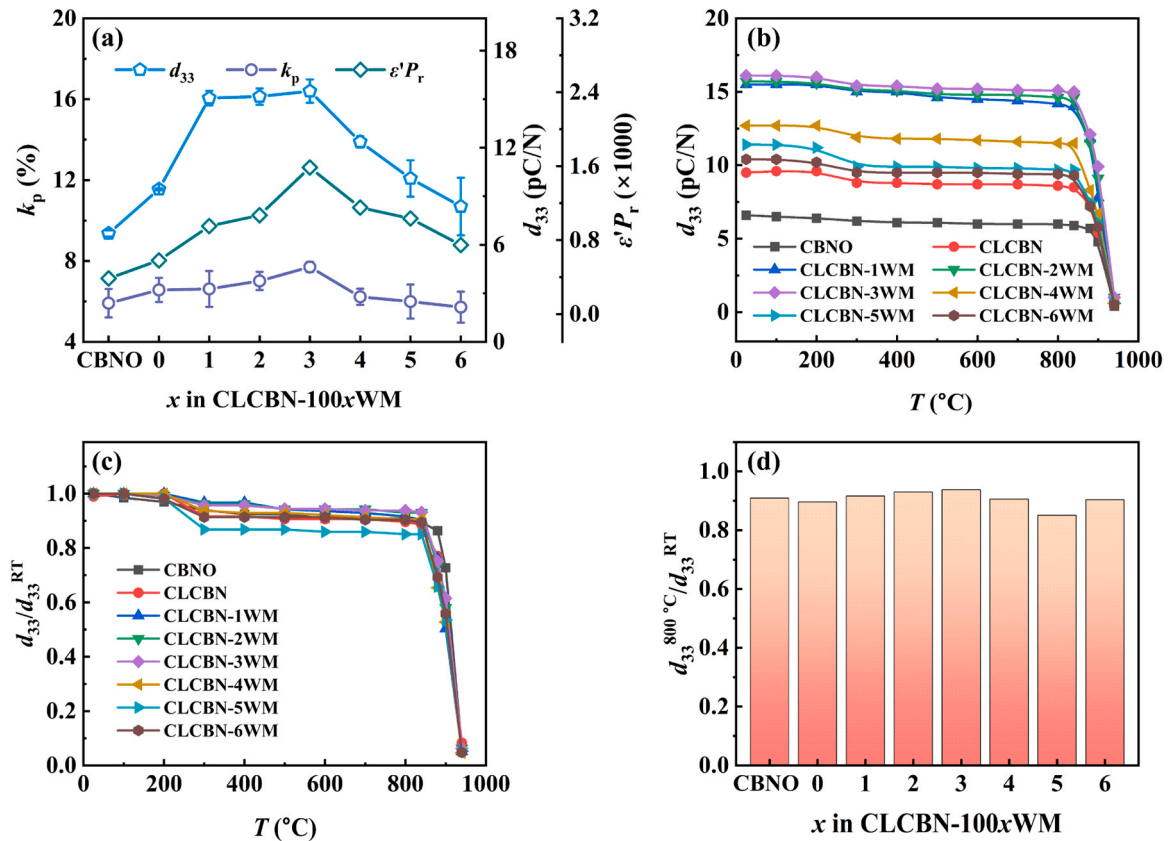


Fig. 6. (a) Composition dependence of d_{33} , k_p , and $\varepsilon'P_r$, (b) Thermal stability of d_{33} , (c) Relationship between normalized d_{33} and annealing temperature, (d) Composition dependence of normalized d_{33} at 800 °C.

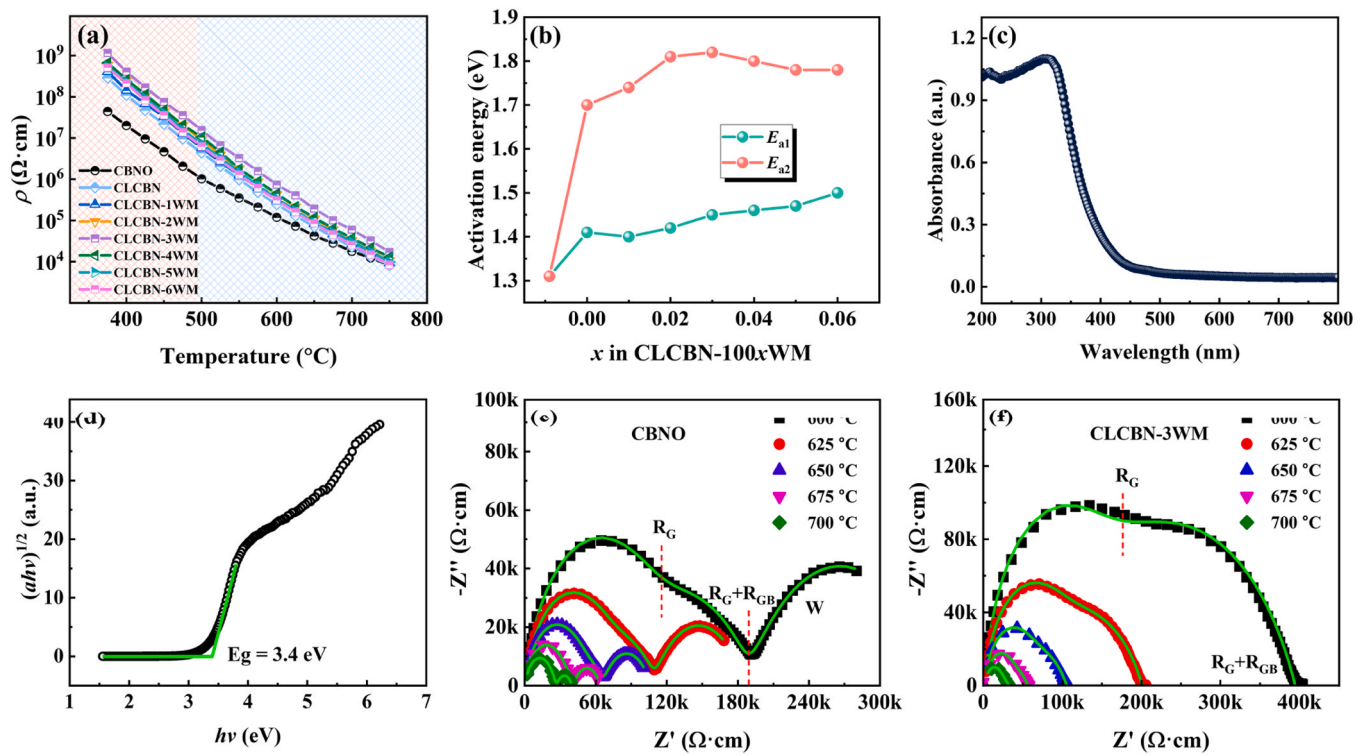


Fig. 7. (a) Relationship between DC resistivity and temperature of all samples, (b) Change of activation energy with composition, UV-Vis diffuse reflectance spectra (c) and band gap width (d) of CBNO ceramics. Impedance spectra of CBNO (e) and CLCBN-3WM (f) ceramics.



The resistivity of the component with $x > 0.03$ decreases slightly due to high levels of W/Mo co-substitution, which introduces excess electrons and reduces the material's insulating performance [15].

To further investigate the conduction mechanism of CLCBN-100xWM ceramics, the activation energy (E_a) for direct current conduction of each sample was calculated using the Arrhenius formula [Eq. (9)], as depicted in Fig. 7(b) and Fig. S8:

$$\rho = \rho_0 \exp(E_a/k_B T) \quad (9)$$

where ρ_0 , k_B , and T represent the pre-exponential factor, the Boltzmann constant ($k_B \sim 8.617 \times 10^{-5}$ eV/K), and the absolute temperature, respectively. The E_a of CBNO ceramics is 1.31 eV, indicating that the conduction mechanism of CBNO is primarily dominated by holes (h') and ionic conduction within the tested temperature range [54,55]. For CLCBN-100xWM ceramics, the E_a cannot be fitted by a single straight line, suggesting different conduction mechanisms within the test range. In the low-temperature range (< 500 °C), E_{a1} ranges between 1.40 and 1.50 eV, which is higher than that of CBNO ceramics. This suggests that the substitution of W/Mo effectively reduces the oxygen vacancy concentration in the material due to the charge compensation mechanism. At higher temperatures (> 500 °C), the calculated E_{a2} ranges from 1.70 to 1.82 eV. Fig. 7(c-d) shows the Ultraviolet-Visible Spectroscopy (UV-Vis) of CBNO samples and the band gap width (E_g) calculated based on the Tauc equation [Eq. (10)]:

$$(\alpha h\nu)^{1/n} = A(h\nu - E_g) \quad (10)$$

where α represents the absorption coefficient, $h\nu$ is the incident photon energy, A is a constant, and the exponent n is a constant related to the type of semiconductor (for direct band gap semiconductors, $n = 1/2$; for indirect band gap semiconductors, $n = 2$) [30,56]. The conduction activation energy E_{a2} of CLCBN-100xWM ceramics at high temperatures is approximately half of the CBNO band gap width ($E_g = 3.4$ eV). This indicates that there is an intrinsic conduction mechanism in the conduction process. Fig. S9 illustrates the impact of oxygen partial pressure (pO_2) on the insulation resistivity of CLCBN-3WM ceramics. The resistivity decreases as the oxygen partial pressure increases from 10^{-4} atm (pure N_2) to 1 atm (pure O_2). This behavior indicates a hole (p -type) conduction mechanism, where holes are generated through the absorption and dissociation of oxygen molecules [Eq. (11)].



To further investigate the conductive mechanism of ceramics under an AC signal, electrochemical impedance spectroscopy was performed on CBNO and CLCBN-3WM samples, as depicted in Fig. 7(e-f). Three semicircles are obtained in the Cole-Cole diagram of the CBNO sample, corresponding to the Warburg effect, grain boundaries, and grains, as the test frequency increases from low to high frequencies [57,58]. Warburg effect reflects the diffusion process of ions near the electrode interface. The parallel R-CPE circuit is utilized to fit the impedance data of CLCBN-3WM ceramics. A distinction exists between the grain resistance and the grain boundary resistance. As the temperature increases, both resistances exhibit a negative correlation with temperature and the difference decreases, as shown in Table S7. In contrast, the Cole-Cole diagram of CLCBN-3WM ceramics exhibits no Warburg effect, indicating that the ionic conducting behavior was suppressed.

4. Conclusions

A novel high-temperature piezoelectric ceramic CLCBN-100xWM was synthesized through the solid-state reaction method. The influence of Li/Ce at A site and Mo/W at B site co-substitution on the morphology, phase structure, and electrical properties of CBNO ceramics was systematically investigated. The co-substitution significantly reduces grain size, enhances spontaneous ferroelectric polarization, and induces a pseudo-tetragonal phase transition in this system. At the same time, the electrons generated by donor doping compensate for the holes, thereby reducing the concentration of oxygen vacancies and enhancing the insulation performance of the ceramic. Notably, CLCBN-3WM ceramic exhibits optimal properties with d_{33} of 16.1 pC/N, T_c around 922 °C, and ρ of 7.4×10^5 Ω -cm at 600 °C, as well as excellent thermal stability which retains 93.7 % (~ 15.1 pC/N) of d_{33} after annealing at 800 °C for 2 h. Therefore, CLCBN-3WM ceramic is a promising material for high-temperature sensors.

CRedit authorship contribution statement

Zhihong Luo: Investigation, Data curation. **Wangxin Li:** Software, Methodology. **Laijun Liu:** Writing – review & editing, Supervision, Funding acquisition. **Shuhang Yu:** Methodology, Formal analysis. **Qiantong Li:** Visualization, Data curation. **Mingmei Lin:** Visualization, Resources. **Junhui Lang:** Visualization, Methodology. **Mingwang Yuan:** Software, Methodology. **Changbai Long:** Supervision. **Dawei Wang:** Visualization, Formal analysis. **Biao Zhang:** Writing – original draft, Investigation, Data curation. **Jianming Deng:** Supervision. **Feng Yan:** Visualization, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jeurceramsoc.2024.117101.

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